

cct College Dublin
Computing • IT • Business

Data Exploration & Preparation

Diploma in Data Analytics for Business – Level 7

CCT College Dublin

Module Introduction

Senior Lecturer: Dr. Muhammad Iqbal*

Introduction

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- **Lecturer:** Dr. Muhammad Iqbal*
- **Experience:** Machine Learning, Data Analytics and Processing, Numerical Modelling & Simulations, Structured & Object-Oriented Programming, Data Structures & Algorithms, Scalable Systems Programming (Python, R, Matlab, etc..).
- **E-mail:** *miqbal@cct.ie*
- **Contact:** Use CCT email address for contact along with your Module and Course names.

Module Information

- **Contact hours:** 4 hours lecture and tutorial (Practical) weekly
- Approximately 3 to 4 hours weekly of independent learning
- **Continuous Assessments**
- **100%** Continuous Assessment
    - **40%** - CA1 (MLO2, MLO3 and MLO4)
    - **60%** - CA2 (MLO1, MLO5 and MLO6)
- **Machine Requirements:** Windows 10/ 11 machine

Module Learning Outcomes

- **MLO 1:** Gather and format data into an actionable resource dependent upon the tools and resources available ensuring compliance with ethical, legal and regulatory standards
- **MLO 2:** Investigate the data using Exploratory Data Analysis (EDA) and discuss the findings with a view to how this will affect decision making.
- **MLO 3:** Implement a data cleaning and preparation plan using appropriate techniques for further analysis
- **MLO 4:** Select and employ appropriate data engineering processes to enhance insight and scope, of the data analysis
- **MLO 5:** Communicate the properties of the data, ensuring consistency and replicability, (Qualitative, Quantitative, Data Type, Feature Type, etc.) using a Data Dictionary
- **MLO 6:** Demonstrate a project cycle, using tools that ensure evidence of verifiable ongoing and regular progress, and incorporating feedback from stakeholders.

Core Resources: Books

- Indicative reading lists and other information resources
- **Core Resources:**

- Ball, R. and Rague, B. (2022) *The beginner's guide to data science*. Cham: Springer.
- Foxwell, H.J. (2020) *Creating good data: A guide to dataset structure and data representation.* Berkeley, CA: Apress.
- McGregor, S.E. (2022) *Practical Python data wrangling and data quality: Getting started with reading, cleaning, and analyzing data*. Sebastopol, CA: O'Reilly Media.
- Narayana, D., Ranjan, S. and Tyagi, N. (2023) *Basic computational techniques for data analysis: An exploration in MS Excel*. 2nd edn. Routledge India.
- AI Publishing (2020) *Data preprocessing with Python for absolute beginners: Step-by-step guide with hands-on projects and exercises*. AI Publishing LLC.

Supplementary Resources

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- Association for Computing Machinery (ACM) (n.d.) *ACM SIGs foster international technical communities within the major areas of computing*. Available at: https://www.acm.org/special-interest-groups.
- Bruce, P.C. and Bruce, A. (2017) *Practical statistics for data scientists: 50 essential concepts*. Sebastopol, CA: O'Reilly.
- Godsey, B. (2017) *Think like a data scientist: Tackle the data science process step-by-step*. Shelter Island, NY: Manning.
- Heatley, L.M. (n.d.) *National Centre for Text Mining — NaCTeM*. NaCTeM. Available at: http://www.nactem.ac.uk/software.php.
- Ilyas, I.F. and Chu, X. (2019) *Data cleaning*. New York, NY: Association for Computing Machinery (ACM Books, 28).
- de Jonge, E. and van der Loo, M. (2018) *Statistical data cleaning with applications in R*. Hoboken, NJ: John Wiley & Sons.
- Matthes, E. (2019) *Python crash course: A hands-on, project-based introduction to programming*. 2nd edn. San Francisco, CA: No Starch Press.
- McCallum, Q.E. (2013) *Bad data handbook: Mapping the world of data problems*. Beijing: O'Reilly.
- Squire, M. (2015) *Clean data*. Birmingham, UK: Packt Publishing. Available at: https://search.ebscohost.com/login.aspx?direct=true&db=nlebk&AN=996715&site=eds-live

CCT Resources

- **Essentials 1 - Study Skills**
    - https://moodle.cct.ie/course/view.php?id=677
- CCT ARC (https://arc.cct.ie/)
- https://www.datacamp.com/community
- For technical support, contact with Mr. Juan Murguey (jmurguey@cct.ie)

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